

Benjamin Tran-Pugh

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TECHNICAL SKILLS

CAD & Analysis: SolidWorks (FEA, PDM, Sheet Metal), Onshape (FEA), Fusion 360, Autodesk Inventor, AutoCAD, GD&T
Software: Python (NumPy, SciPy, Matplotlib), C/C++ (ESP32), MATLAB, Linux (Raspberry Pi, Jetson Orin Nano), Git
Other: 3D Printing (PLA, PETG, TPU, ABS, PA6-CF), DFM/DFA, Machining, Technical Drawings, Microsoft Suite, Soldering

EXPERIENCE

Mechanical Engineering Intern

Feb 2026 – Aug 2026

Human Computer Lab (SPC '25)

Toronto, ON

- Owned mechanical and actuation development of LeLamp, an animatronic robotics product across three actuator generations, leading to improvement of actuation noise by 20dB and thermal output of actuators to fall 41.6%
- Drove head design by simulating heat buildup, camera obstruction, and light diffusion from first principles, accurately predicting brightness and thermal performance, leading to increase in light by 416% and reduction in heat by 46.1%
- Led R&D efforts in building animated robots from scratch, integrating impedance control, quiet operating noise, and led testing to determine safety margins on torque, overheating, noise, and vibrations against simulation results
- Designed and hardware-validated a spring-based gravity compensation system to eliminate static actuator thermal load by 92.3%, deriving joint balance conditions from Lagrangian mechanics and verifying against published literature
- Designed mechanisms such as antagonistic pairs, capstan drive, and full/partial spring gravity compensation by running through calculations, simulating physics, testing real-world results against simulations, and iterating
- Identified root causes of motor noise from first principles and reduced contact-driven vibrational resonance by 15dB

Mechanical Team Member

Sep 2025 – Apr 2026

Waterloo Aerial Robotics Group

Waterloo, ON

- Designed and fabricated a sub-60 g payload-release mechanism for the Valkyrie competition drone, securing objects of varying size and mass in flight and releasing them at a small target with snag-free, propeller-safe routing
- Designed iterations of empennage structures for Helios, a fixed-wing aircraft evolving toward VTOL capability
- Managed design data in SolidWorks PDM across a multi-hundred-member team, integrating mechanical designs with electrical and autonomy subsystems, to help team win the Pip Rudkin Perseverance Award at AEAC 2026

PROJECTS

Active Suspension Model

Python, ISO8608, Vehicle Dynamics, Vibrations, SolidWorks

- Simulated suspension system of a quarter-car rig by deriving EOM, researching real parameters, implementing skyhook control, generated ISO8608 road profiles, and writing scripts using NumPy, SciPy, and Matplotlib
- Simulated an improvement to ride comfort by 9.6% on ISO8608 class C type roads, and 57.1% during resonance
- Physically built out full quarter-car rig using SolidWorks and tested dropdown, hold and release, and sinusoidal roads

Double Pendulum Chaos Simulation

Python, SciPy, NumPy, Matplotlib

- Derived the full nonlinear equations of motion from first principles by hand from the Euler-Lagrange equation, obtaining closed-form angular accelerations via Cramer's rule, which allowed physics simulations to run
- Built a numerical simulation using python (NumPy, Matplotlib, and SciPy) demonstrating sensitive dependence and chaotic motion arising from variations in initial conditions as small as $1e-9$ rad
- Computed and visualized a time-to-flip fractal across the initial-condition space using NumPy, SciPy, and Matplotlib

4-DOF Robotic Arm 2.0

SolidWorks, ESP32, C++

- Modeled a 50+ part assembly and validated 8 structural components with FEA against load requirements
- Implemented 4-DOF inverse kinematics, trajectory smoothing, and a web-based control interface using C/C++ running directly on the ESP32 for the built-in WiFi and Bluetooth capabilities
- Wrote Python and MATLAB simulations to verify joint limits and workspace before hardware runs, preventing clashing, tipping, and overheating

EDUCATION

University of Waterloo

Sep 2025 – Jun 2030

Candidate for BAsC, Mechanical Engineering – Cumulative GPA: 3.83/4.0

Waterloo, ON